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EXPERIMENT – 25

SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF ENCODED OUTPUT BIT SEQUENCES USING MATLAB

PROGRAM:

```
clc;
clear;
close all;
% Input Bits
data = randi([0 1],20,1);
% Convolutional Encoder (Rate 1/2, Constraint Length = 3)
trellis = poly2trellis(3,[7 5]);
encoded = convenc(data,trellis);
% Display
disp('Input Bits:');
disp(data');
disp('Encoded Bits:');
disp(encoded');
% Plot
figure;
subplot(2,1,1)
stem(data,'filled')
title('Input Bit Sequence')
xlabel('Bit Index')
ylabel('Bit Value')
grid on
subplot(2,1,2)
stem(encoded,'filled')
```

```
title('Convolutionally Encoded Bit Sequence')
xlabel('Bit Index')
ylabel('Bit Value')
grid on
```

OUTPUT:

